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1. An acute-angled triangle  $ABC$  has orthocentre  $H$ . The circle passing through  $H$  with centre the midpoint of  $BC$  intersects  $BC$  at  $A_1$  and  $A_2$ . Similarly define points  $B_1, B_2, C_1, C_2$ . Show that  $A_1, A_2, B_1, B_2, C_1, C_2$  lie on a circle.

2. (a) Prove that

$$\frac{x^2}{(x-1)^2} + \frac{y^2}{(y-1)^2} + \frac{z^2}{(z-1)^2} \geq 1$$

for all real numbers  $x, y, z \neq 1$  satisfying  $xyz = 1$ .

- (b) Prove that equality holds for infinitely many rational triples  $(x, y, z)$ .

3. Prove that there exist infinitely many positive integers  $n$  such that  $n^2 + 1$  has a prime divisor greater than  $2n + \sqrt{2n}$ .

4. Find all functions  $f : (0, \infty) \rightarrow (0, \infty)$  such that

$$\frac{f(w)^2 + f(x)^2}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2}$$

for all positive real numbers  $w, x, y, z$  satisfying  $wx = yz$ .

5. Let  $n$  and  $k$  be positive integers with  $k \geq n$  and  $k - n$  even. There are  $2n$  lamps, initially all off. At each step one lamp is switched.

Let  $N$  be the number of sequences of  $k$  steps resulting in lamps 1 through  $n$  on and lamps  $n + 1$  through  $2n$  off.

Let  $M$  be the number of such sequences where lamps  $n + 1$  through  $2n$  are never switched on. Determine  $N/M$ .

6. Let  $ABCD$  be a convex quadrilateral with  $|BA| \neq |BC|$ . Let  $\omega_1$  and  $\omega_2$  be the incircles of triangles  $ABC$  and  $ADC$ . Suppose a circle  $\omega$  is tangent to ray  $BA$  beyond  $A$  and ray  $BC$  beyond  $C$ , and also tangent to lines  $AD$  and  $CD$ . Prove that the common external tangents of  $\omega_1$  and  $\omega_2$  intersect on  $\omega$ .